

$$\begin{aligned}
& ((\lambda a. (\lambda b. b \ b) (\lambda b. b \ b)) \ b) ((\lambda c. (c \ b)) (\lambda a. a)) \\
= & ((\lambda a. \Omega) \ b) ((\lambda c. (c \ b)) \ I) \\
\rightarrow_{\beta} & \Omega ((\lambda c. (c \ b)) \ I) \\
\rightarrow_{\beta} & \Omega (I \ b) \\
\rightarrow_{\beta} & \Omega \ b
\end{aligned}$$